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EXPERIMENT 20

Exp No:20 Experiment to understand K-means clustering algorithm for a given data set.

AIM:
To understand K-means clustering algorithm for a given data set.

DESCRIPTION:
Understand the K-means clustering algorithm for the dataset given.

CODE:

```
 import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

df = pd.read_csv('SuperMarket Analysis-selected-columns.csv')
df.info()

print(df.head())

sns.pairplot(df, vars=['Unit price', 'Quantity', 'Tax 5%', 'Sales'])
plt.show()

features = df.iloc[:, [6, 9]].values
print(features[:5])
from sklearn.cluster import KMeans

model = KMeans(n_clusters=5, n_init=10, random_state=42)
model.fit(features)
print(model)
Final = df.iloc[:, [6, 9]].copy()
Final['label'] = model.predict(features)
print(Final.head())
```

```
print(Final['label'].values)

print(Final['label'].value_counts().sort_index())

sns.set_style("whitegrid")
sns.FacetGrid(Final, hue="label", height=8) \
.map(plt.scatter, "Unit price", "Sales") \
.add_legend()
plt.show()
features_el = df.iloc[:, [6, 7, 9]].values

from sklearn.cluster import KMeans

wcss = []

for i in range(1, 10):
    model_el = KMeans(n_clusters=i, n_init=10, random_state=42)
    model_el.fit(features_el)
    wcss.append(model_el.inertia_)

for i in range(1, 10):
    print("Number of clusters:", i, "WCSS:", wcss[i-1])

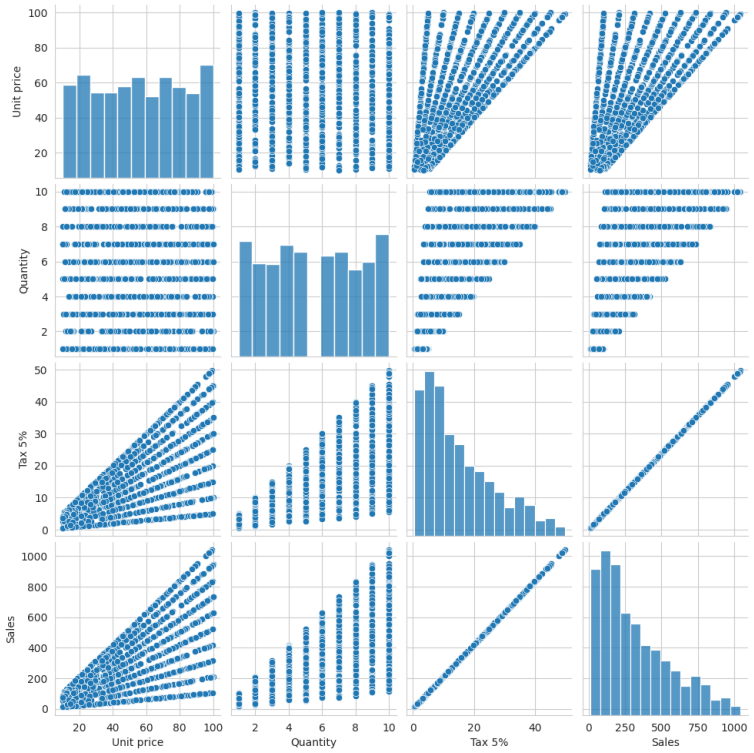
plt.plot(range(1, 10), wcss)
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Invoice ID      1000 non-null   object
 1   Branch          1000 non-null   object
 2   City            1000 non-null   object
 3   Customer type   1000 non-null   object
 4   Gender          1000 non-null   object
 5   Product line    1000 non-null   object
 6   Unit price      1000 non-null   float64
 7   Quantity        1000 non-null   int64
 8   Tax 5%          1000 non-null   float64
 9   Sales           1000 non-null   float64
dtypes: float64(3), int64(1), object(6)
memory usage: 78.3+ KB
```

	Invoice ID	Branch	City	Customer type	Gender
0	750-67-8428	Alex	Yangon	Member	Female
1	226-31-3081	Giza	Naypyitaw	Normal	Female
2	631-41-3108	Alex	Yangon	Normal	Female
3	123-19-1176	Alex	Yangon	Member	Female
4	373-73-7910	Alex	Yangon	Member	Female

	Product line	Unit price	Quantity	Tax 5%	Sales
0	Health and beauty	74.69	7	26.1415	548.9715
1	Electronic accessories	15.28	5	3.8200	80.2200
2	Home and lifestyle	46.33	7	16.2155	340.5255
3	Health and beauty	58.22	8	23.2880	489.0480
4	Sports and travel	86.31	7	30.2085	634.3785



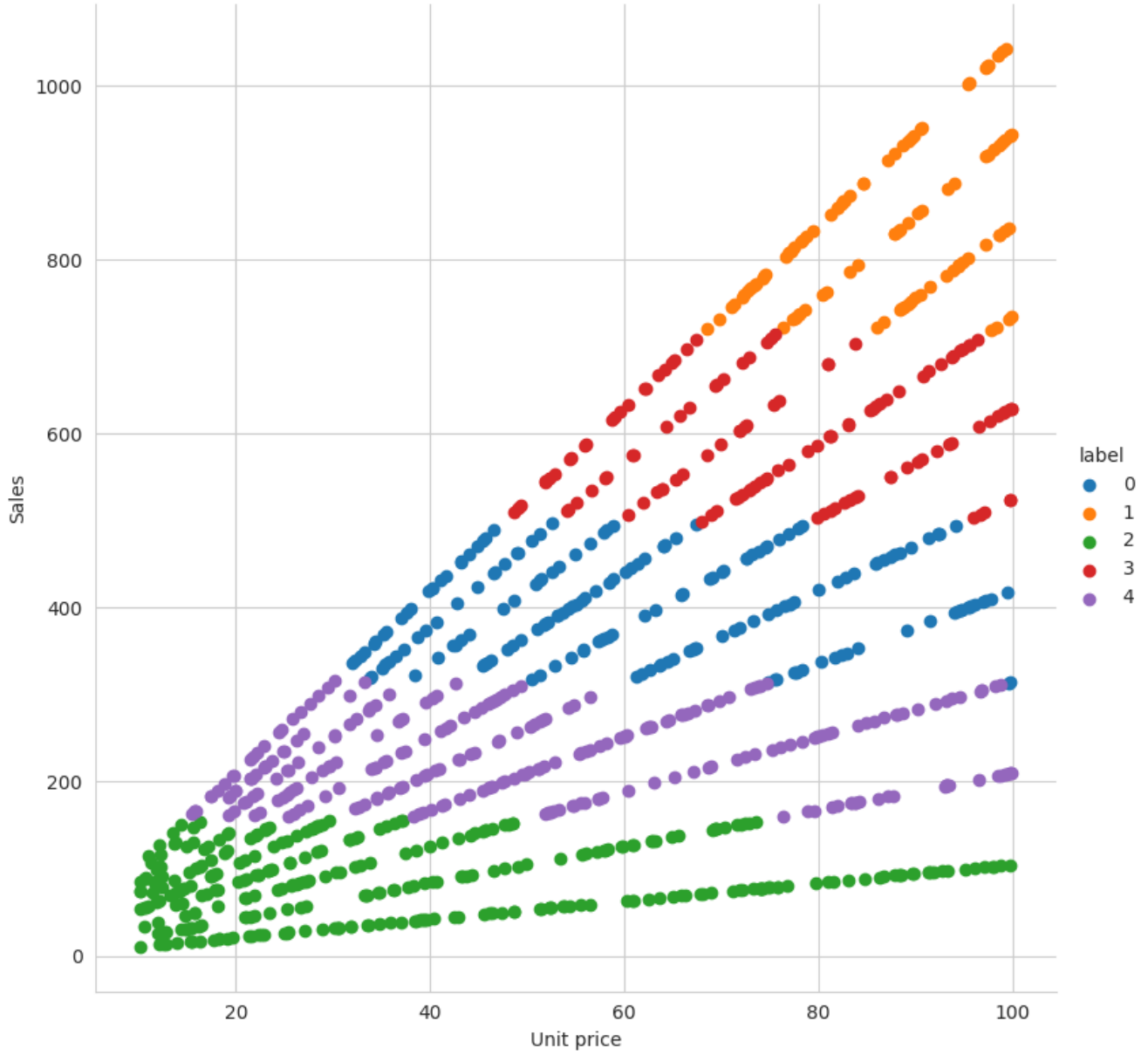
```
[[ 74.69   548.9715]
 [ 15.28    80.22  ]
 [ 46.33   340.5255]
 [ 58.22   489.048 ]
 [ 86.31   634.3785]]
KMeans(n_clusters=5, n_init=10, random_state=42)
```

	Unit price	Sales	label
0	74.69	548.9715	3
1	15.28	80.2200	2
2	46.33	340.5255	0
3	58.22	489.0480	0
4	86.31	634.3785	3

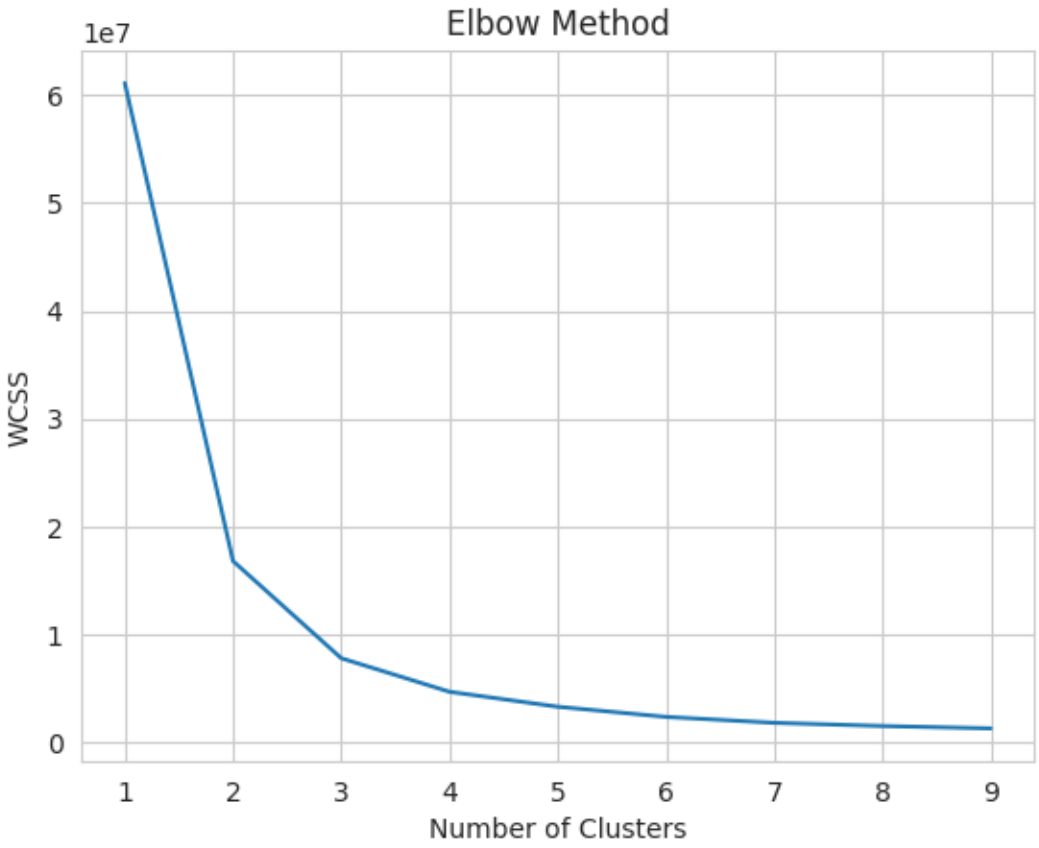
```
[3 2 0 0 3 3 0 1 2 4 2 2 4 0 1 3 3 0 4 2 0 4 2 4 4 0 2 4 0 4 0 1 3 4 0 2 0
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 2 4 2 2 4 4 2 4 1 4 3 4 4 2 4 4 2 2 4 4 1 1 3 0 2 3 1 3 4 1 2 4 2 2 1 2 2
 3]
```

label	
0	185
1	101
2	322
3	127
4	265

Name: count, dtype: int64




```
Number of clusters: 1 WCSS: 61108939.68566262
Number of clusters: 2 WCSS: 16822834.499406073
Number of clusters: 3 WCSS: 7817933.856244104
Number of clusters: 4 WCSS: 4709311.546266038
Number of clusters: 5 WCSS: 3324297.95296186
Number of clusters: 6 WCSS: 2383449.0975037576
Number of clusters: 7 WCSS: 1834458.3940780417
Number of clusters: 8 WCSS: 1538357.2999423782
Number of clusters: 9 WCSS: 1318906.3851972432
```



RESULT:

Thus, the K-means clustering algorithm was implemented successfully to group supermarket sales data into clusters. The clusters were visualized using scatter plots, and the optimal number of clusters was analyzed using the Elbow Method.